Set Theory

2 INTRODUCTION:

A set is an unordered collection of objects and is determined by the objects it contains. Before 19th century it was uncommon to think of sets as completed objects in their own right. Mathematicians were familiar with properties such as being a rational number or irrational number, but it was rare to think of collection of rational numbers as itself an object. There were exceptions too. From the time of Euclid, Mathematicians were used to think of geometric objects such as lines, planes, spheres etc which we today identify with their sets of points. Set theory is an important language and tool for reasoning. It is a basis for Mathematics. Most of the Mathematics can be formalised in set theory.

Set theory is a useful tool for formalising and reasoning about computation and the objects of computation. It is indivisible from logic where computer science has its roots. It is likely to continue to be a source of fundamental ideas in computer science from theory to practice. Computer science, being a science of the artificial, has had many of its constructs and ideas inspired by set theory. The strong tradition, universality and neutrality of set theory make it firm common ground on which to provide unification between seemingly desparate areas and notations of computer science. Set theory is likely to be around long after most present day programming languages have faded to think abstractly. It provides us with a foundation on which to build a firm understanding and analysis of the new ideas in computer science that we will meet.

2.1 Sets and Elements :

A set is usually defined as an unordered collection of well defined objects or things. The objects or things are called elements or members of the set. For example, the collection of all states in India, the collection of all integers, the collection of fruits in the market, the collection of all the students in a college etc are all sets.

Notation :

The symbols $\in$ (read as belongs to) is used to denote the membership in a set. The symbol $\notin$ (read as does not belong to) denotes that a particular element is not a member of a set. Sets are usually denoted by capital letters.

Let $A = \{a, b, c, d, e\}$

Then $c \in A$, $d \in A$, $x \notin A$.

Scanned by TapScanner

2.2 Specifying sets :

There are two different ways of specifying sets.

(i) Listing Method or Roaster Method :

In this method all the elements of the set are separated by commas and the elements are unique.

For example, A = {1, 6, 4, 3}

(ii) Set Builder Method or Ruler Method :

In this method all the elements of the set are denoted by x where x satisfies certain property.

For example, let A = { x | x has the property P }

Here the set A denotes all the elements x such that x has the property P.

For example, let A = {x | x ∈ R} i.e. A = { ....., - 3, - 2, - 1, 0, 1, 2, 3, ...}

where R is the set of real numbers

D = {y | 0 ≤ y ≤ 3, y ∈ Z}

where Z is the set of integers

i.e. D = {0, 1, 2, 3}.

2.3 Types of sets :

(i) Finite Set :

A set A is said to be finite, if it contains finite number of elements. Finite sets are also called **numerable sets**.

For example, A = {0, 1, 2, 3}, B = {x | 0 ≤ x ≤ 2, x ∈ Z} are finite sets.

(ii) Infinite Set :

A set A is said to be infinite, if it contains infinite number of elements.

For example, A = {0,1, 2, 3, ... }, B = {x | x is a real number} are infinite sets.

(iii) Singleton Set :

A set A is said to be singleton set, if it contains only one element.

For example, A = {0}, D = {7} are singleton sets.

(iv) Null Set or Empty Set :

A set A is said to be null set or empty set, if it contains no elements. It is denoted by ϕ where ϕ = { }.

Scanned by TapScanner

(v) Subset :

A set A is said to be a subset of the set B written as $A \subseteq B$, if and only if every element of A is also an element of B. We also say that A is contained in B or B contains A ($B \supseteq A$).

Symbolically, If $x \in A \Rightarrow x \in B$ then $A \subseteq B$

For example, let $A = \{1, 2, 3, 4\}$, $B = \{1, 2, 3, 4, 5, 6\}$

Here $A \subseteq B$ because every element of A is also an element of B.

If $A \subseteq B$, then it may be possible that $A = B$. This is the reason why we write $A \subseteq B$.

Note :

(i) Every set is a subset of itself.

i.e. $A \subseteq A, B \subseteq B, C \subseteq C$ etc

(ii) Empty set ϕ is the subset of all sets.

(vi) Proper Subset :

A set A is called a proper subset of the set B if (i) A is the subset of B and (ii) B is not a subset of A. In other words, A is said to be a proper subset of B if every element of A belongs to the set B, but there is atleast one element of B which does not belong to A. If A is a proper subset of B, then it is denoted by $A \subset B$.

For example, let $A = \{x, y, z, w\}$; $B = \{x, y, z, w, r, s, t\}$

Here $A \subseteq B$ but $B \not\subseteq A$

$\therefore A \subset B$

(vii) Equal Sets :

Two non-empty sets A and B are said to be equal if

(i) $A \subseteq B$ (ii) $B \subseteq A$

i.e. every element of A is an element of B and every element of B is an element of A.

For example, if $A = \{1, 2, 3, 4\}$ and $B = \{3, 2, 1, 4\}$, then $A = B$

(viii) Universal Set :

All the sets are likely to be considered as a subset of a particular set. This set is called universal set and is denoted by U.

(ix) Power Set :

For a given non-empty set A, the collection of all subsets of A is called the power set of A and is denoted by P (A). For example, let $A = \{1, 2, 3\}$.

Scanned by TapScanner

The subsets of A are ϕ, {1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3}, {1, 2, 3}.

$\therefore$ P (A) = {ϕ, {1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3}, {1, 2, 3} }

Note :

If a set A contains n elements, then P (A) contains 2^n elements.

(x) Super Set :

If A is a subset of B, then B is called superset of A.

For example, let A = {a, b, c}, B = {a, c, b, d}

Here $A \subseteq B$

$\therefore$ B is a superset of A

i.e. $B \supseteq A$

Note :

Universal set is the superset of all sets.

i.e. if A, B, C are any sets, then $U \supseteq A$, $U \supseteq B$, $U \supseteq C$ etc.

(xi) Disjoint Sets :

Two non-empty sets A and B are said to be disjoint if they have no element in common.

For example, let A = {1, 2, 3}, B = {- 3, 4, 6}

Here A and B have no common element

$\therefore$ A and B are disjoint sets.

(xii) Comparable and Non - Comparable Sets :

Let A and B be any two sets. Then A and B are said to be comparable if $A \subset B$ or $B \subset A$. Also A and B are said to be non-compable if $A \not\subset B$ and $B \not\subset A$.

(xiii) Multiset and Multiplicity :

Multiset is a collection of objects, where the objects may not be distinct.

Let A be any set and $x \in A$. Then the multiplicity of x is defined to be the number of times the element x appears in the set A.

For example, A = {1, 1, 2, 3, 3, 4, 5} is a multiset.

Here multiplicity of 1 is 2,

multiplicity of 3 is 2 etc.

Also B = {x, x, x, y, z, z, w, w, w, w} is a multiset.

Here multiplicity of x is 3

multiplicity of y is 1

multiplicity of z is 2

multiplicity of w is 4

Scanned by TapScanner

Note :

If A and B are multisets, then $A \cup B$ and $A \cap B$ are also multisets.

Let $A = \{1, 1, 2, 2, 3, 4, 4\}$ and $B = \{1, 1, 2, 3, 3, 4, 4\}$

Then

$A \cup B = \{1, 1, 2, 2, 3, 3, 4, 4\}$ and

$A \cap B = \{1, 1, 2, 3, 4, 4\}$, which are clearly multisets.

(xiv) Partition of a Set :

Let A be any non-empty set. Then A can be partitioned into several subsets such that each element of A is exactly in one of the subsets of A.

For example, let $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$

Then $\{1, 2, 3\}, \{4\}, \{5, 6\}, \{7, 8\}$ are the different partitions of A.

2.4 Properties of Sets :

(1) If A is any set, then $A \subseteq A$

(2) If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$ where A, B, C are sets.

2.5 Operations on Sets :

(i) Union of Sets :

Let A and B be any two non-empty sets. Then the union of A and B denoted by $A \cup B$, is a set of all elements which belong to A or B or both.

For example, let $A = \{1, 3, 5\}$, $B = \{1, 2, 4, 6\}$

Then $A \cup B = \{1, 2, 3, 4, 5, 6\}$

(ii) Intersection of Sets :

Let A and B be any two non-empty sets. Then the intersection of A and B denoted by $A \cap B$, is a set of all elements which belong to both A and B.

For example, let $A = \{a, b, d, g\}$, $B = \{b, d, f, e\}$

Then $A \cap B = \{b, d\}$

(iii) Complement of a Set :

Let A be any non-empty set. Then $A \subseteq U$, where U is the universal set. Then the complement of A is denoted by A^C or A' or $\overline{A}$ and is defined as the collection of all elements which belong to U but does not belong to A.

i.e. $A' (A^C \text{ or } \overline{A}) = \{x \mid x \in U, x \notin A\}$

For example, let $A = \{1, 2, 3, 4\}$, $U = \{1, 2, 3, 4, 5, 6, ...\}$

Then

$A' (A^C \text{ or } \overline{A}) = \{5, 6, 7, 8, ...\}$

Scanned by TapScanner

Note :

(i) The empty set ϕ is the complement of universal set U.

(ii) The universal set U is the complement of empty set ϕ.

(iv) Relative Complement or Difference of Sets :

Let A and B be the subsets of the universal set U. Then the difference of A and B, denoted by A - B, is the set containing all the elements of A but not containing the elements of B.

i.e. $A - B = \{x \mid x \in A, x \notin B\}$

For example, let $A = \{p, q, r, s\}$, $B = \{p, a, s, t, y\}$

Then

$A - B = \{q, r\}$, $B - A = \{a, t, y\}$

(v) Symmetric Difference :

The symmetric difference of two non-empty sets A and B is denoted by $A \Delta B$ or $A \oplus B$ and is defined as the union of A - B and B - A.

i.e. $A \Delta B\ (A \oplus B) = (A - B) \cup (B - A) = (A \cap B^1) \cup (B \cap A^1)$

For example, let $A = \{a, b, c\}$, $B = \{b, c, d\}$

Then $A - B = \{a\}$, $B - A = \{d\}$

$\therefore$ $A \Delta B\ (\text{or } A \oplus B) = (A - B) \cup (B - A) = \{a, d\}$

2.6 Principle of Duality :

The principle of duality states that any established result involving sets and complements and operations of union and intersection gives a corresponding dual result by replacing U by ϕ and $\cup$ by $\cap$ and vice - versa.

For example, consider $A \cup \overline{A} = U$

Then By principle of duality, $A \cap \overline{A} = \phi$

2.7 Laws of Algebra on sets :

(1) $A \cup A = A$

(2) $A \cap A = A$

(3) $A \cup \phi = A$

(4) $A \cup U = U$

Scanned by TapScanner

(5) Commutative Laws : (i) $A \cup B = B \cup A$

(ii) $A \cap B = B \cap A$

(6) Associative Laws : (i) $(A \cup B) \cup C = A \cup (B \cup C)$

(ii) $(A \cap B) \cap C = A \cap (B \cap C)$

(7) Distributive Laws : (i) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

(ii) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(8) Absorption Laws : (i) $(A \cap B) \cup A = A$

(ii) $(A \cup B) \cap A = A$

Proof : **(1) To prove : $A \cup A = A$**

We prove that $A \cup A \subseteq A$ and $A \subseteq A \cup A$

Let $x \in A \cup A$

i.e. $x \in A$ or $x \in A$

i.e. $x \in A$

$\therefore \; x \in A \cup A \Rightarrow x \in A$

$\therefore \; A \cup A \subseteq A$ ------ (a)

Let $x \in A$

i.e. $x \in A$ or $x \in A$

i.e. $x \in A \cup A$

$x \in A \Rightarrow x \in A \cup A$

$\therefore \; A \subseteq A \cup A$ -------(b)

From (a) and (b) we get $A \cup A = A$

(2) To prove : $A \cap A = A$

We prove that $A \cap A \subseteq A$ and $A \subseteq A \cap A$

Let $x \in A \cap A$

i.e. $x \in A$ and $x \in A$

i.e. $x \in A$

$\therefore \; x \in A \cap A \Rightarrow x \in A$

$\therefore \; A \cap A \subseteq A$ ------ (a)

Let $x \in A$

i.e. $x \in A$ and $x \in A$

i.e. $x \in A \cap A$

$\therefore \; x \in A \Rightarrow x \in A \cap A$

Scanned by TapScanner

$\therefore A \subseteq A \cap A$ -------- (b)

From (a) and (b), we get $A \cap A = A$

(3) To prove : $A \cup \phi = A$

We prove that $A \cup \phi \subseteq A$ and $A \subseteq A \cup \phi$

Let $x \in A \cup \phi$

i.e. $x \in A$ or $x \in \phi$

i.e. $x \in A$

$\therefore x \in A \cup \phi \Rightarrow x \in A$

$\therefore A \cup \phi \subseteq A$ ------- (a)

Clearly $A \subseteq A \cup \phi$ ------ (b)

From (a) and (b), we get $A \cup \phi = A$

(4) To prove : $A \cup U = U$

We prove that $A \cup U \subseteq U$ and $U \subseteq A \cup U$

We know that every set is a subset of the universal set

i.e. $(A \cup U) \subseteq U$

Also $U \subseteq (A \cup U)$

$\therefore A \cup U = U$

(5) (i) To prove : $A \cup B = B \cup A$

We prove that $A \cup B \subseteq B \cup A$ and $B \cup A \subseteq A \cup B$

Let $x \in A \cup B$

i.e. $x \in A$ or $x \in B$

i.e. $x \in B$ or $x \in A$

i.e. $x \in B \cup A$

$\therefore x \in A \cup B \Rightarrow x \in B \cup A$

$\therefore A \cup B \subseteq B \cup A$ -------- (a)

Let $x \in B \cup A$

i.e. $x \in B$ or $x \in A$

i.e. $x \in A$ or $x \in B$

i.e. $x \in A \cup B$

$\therefore x \in B \cup A \Rightarrow x \in A \cup B$

Scanned by TapScanner

$\therefore$ $B \cup A \subseteq A \cup B$ -------- (b)

From (a) and (b), we get $A \cup B = B \cup A$

(ii) To prove : $A \cap B = B \cap A$

We prove that $A \cap B \subseteq B \cap A$ and $B \cap A \subseteq A \cap B$

Let $x \in A \cap B$

i.e. $x \in A$ and $x \in B$

i.e. $x \in B$ and $x \in A$

i.e. $x \in B \cap A$

$\therefore$ $x \in A \cap B \Rightarrow x \in B \cap A$

$\therefore$ $A \cap B \subseteq B \cap A$ -------- (a)

Let $x \in B \cap A$

i.e. $x \in B$ and $x \in A$

i.e. $x \in A$ and $x \in B$

i.e. $x \in A \cap B$

$\therefore$ $x \in B \cap A \Rightarrow x \in A \cap B$

$B \cap A \subseteq A \cap B$ -------- (b)

From (a) and (b) , we get $A \cap B = B \cap A$.

(6) (i) To prove : $(A \cup B) \cup C = A \cup (B \cup C)$ **(W-09)**

We prove that $(A \cup B) \cup C \subseteq A \cup (B \cup C)$ and

$A \cup (B \cup C) \subseteq (A \cup B) \cup C$

Let $x \in (A \cup B) \cup C$

i.e. $x \in (A \cup B)$ or $x \in C$

i.e. $(x \in A$ or $x \in B)$ or $x \in C$

i.e. $x \in A$ or $(x \in B$ or $x \in C)$

i.e. $x \in A$ or $(x \in B \cup C)$

i.e. $x \in A' \cup (B \cup C)$

$\therefore$ $x \in (A \cup B) \cup C \Rightarrow x \in A \cup (B \cup C)$

$\therefore$ $(A \cup B) \cup C \subseteq A \cup (B \cup C)$ -------- (a)

Let $x \in A \cup (B \cup C)$

i.e. $x \in A$ or $x \in (B \cup C)$

i.e. $x \in A$ or $(x \in B$ or $x \in C)$

Scanned by TapScanner

i.e. $(x \in A \text{ or } x \in B) \text{ or } x \in C$

i.e. $x \in (A \cup B) \text{ or } x \in C$

i.e. $x \in (A \cup B) \cup C$

$\therefore \quad x \in A \cup (B \cup C) \Rightarrow x \in (A \cup B) \cup C$

$\therefore \quad A \cup (B \cup C) \subseteq (A \cup B) \cup C$ ---------- (b)

From (a) and (b), we get $(A \cup B) \cup C = A \cup (B \cup C)$

(ii) To prove : $(A \cap B) \cap C = A \cap (B \cap C)$

We prove that $(A \cap B) \cap C \subseteq A \cap (B \cap C)$ and

$$A \cap (B \cap C) \subseteq (A \cap B) \cap C$$

Let $x \in (A \cap B) \cap C$

i.e. $x \in (A \cap B) \text{ and } x \in C$

i.e. $(x \in A \text{ and } x \in B) \text{ and } x \in C$

i.e. $x \in A \text{ and } (x \in B \text{ and } x \in C)$

i.e $x \in A \cap (B \cap C)$

$\therefore \quad x \in (A \cap B) \cap C \Rightarrow x \in A \cap (B \cap C)$

$\therefore \quad (A \cap B) \cap C \subseteq A \cap (B \cap C)$ ---------- (a)

Let $x \in A \cap (B \cap C)$

i.e. $x \in A \text{ and } x \in (B \cap C)$

i.e. $x \in A \text{ and } (x \in B \text{ and } x \in C)$

i.e. $(x \in A \text{ and } x \in B) \text{ and } x \in C$

i.e. $x \in (A \cap B) \text{ and } x \in C$

i.e. $x \in (A \cap B) \cap C$

$\therefore \quad x \in A \cap (B \cap C) \Rightarrow x \in (A \cap B) \cap C$

$\therefore \quad A \cap (B \cap C) \subseteq (A \cap B) \cap C$ ---------- (b)

From (a) and (b) we get $(A \cap B) \cap C = A \cap (B \cap C)$

(7) To prove : (i) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

We prove that $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$ and

$$(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$$

Let $x \in A \cap (B \cup C)$

i.e. $x \in A \text{ and } x \in (B \cup C)$

i.e. $x \in A \text{ and } (x \in B \text{ or } x \in C)$

i.e. $(x \in A \text{ and } x \in B) \text{ or } (x \in A \text{ and } x \in C)$

Scanned by TapScanner

i.e. $x \in (A \cap B)$ or $x \in (A \cap C)$

i.e. $x \in (A \cap B) \cup (A \cap C)$

$\therefore$ $x \in A \cap (B \cup C) \Rightarrow x \in (A \cap B) \cup (A \cap C)$

$\therefore$ $A \cap (B \cup C) \subseteq (A \cap B) \cup (A \cap C)$ ---------- (a)

Let $x \in (A \cap B) \cup (A \cap C)$

i.e. $x \in (A \cap B)$ or $x \in (A \cap C)$

i.e. ($x \in A$ and $x \in B$) or ($x \in A$ and $x \in C$)

i.e. $x \in A$ and ($x \in B$ or $x \in C$)

i.e. $x \in A$ and $x \in (B \cup C)$

i.e. $x \in A \cap (B \cup C)$

$\therefore$ $x \in (A \cap B) \cup (A \cap C) \Rightarrow x \in A \cap (B \cup C)$

$\therefore$ $(A \cap B) \cup (A \cap C) \subseteq A \cap (B \cup C)$ ---------- (b)

From (a) and (b), we get $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

(ii) To prove : $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

This result can be easily proved using the above result (i).

(8) (i) To prove : $(A \cap B) \cup A = A$

LHS = $(A \cap B) \cup A$

= $(A \cap B) \cup (A \cap U)$ ($\because A \cup U = A$)

= $A \cap (B \cup U)$ (by distributive law)

= $A \cap U$ ($\because B \cup U = U$)

= A

= RHS

(ii) To prove : $(A \cup B) \cap A = A$

LHS = $(A \cup B) \cap A$

= $A \cap (A \cup B)$ (by commutative law)

= $A \cap (B \cup A)$

= $(A \cap B) \cup (A \cap A)$ (by distribution law)

= $(A \cap B) \cup A$ (by (2))

= A (by 8 (i))

= RHS

$\therefore$ $(A \cup B) \cap A = A$

Scanned by TapScanner

De Morgan's Laws : (S-07, 09)

(1) $(A \cup B)' = A' \cap B'$ (2) $(A \cap B)' = A' \cup B'$

Proof : (1) **To prove :** $(A \cup B)' = A' \cap B'$

We prove that $(A \cup B)' \subseteq A' \cap B'$ and $A' \cap B' \subseteq (A \cup B)'$

Let $x \in (A \cup B)'$

i.e. $x \notin A \cup B$

i.e. $x \notin A$ or $x \notin B$

i.e. $x \in A'$ and $x \in B'$

i.e. $x \in A' \cap B'$

$\therefore x \in (A \cup B)' \Rightarrow x \in A' \cap B'$

$\therefore (A \cup B)' \subseteq A' \cap B'$ --------- (a)

Let $x \in A' \cap B'$

i.e. $x \in A'$ and $x \in B'$

i.e. $x \notin A$ or $x \notin B$

i.e. $x \notin (A \cup B)$

i.e. $x \in (A \cup B)'$

$\therefore x \in A' \cap B' \Rightarrow x \in (A \cup B)'$

$\therefore A' \cap B' \subseteq (A \cup B)'$ --------- (b)

From (a) and (b), we get $(A \cup B)' = A' \cap B'$

(2) To prove : $(A \cap B)' = A' \cup B'$

We prove that $(A \cap B)' \subseteq A' \cup B'$ and $A' \cup B' \subseteq (A \cap B)'$

Let $x \in (A \cap B)'$

i.e. $x \notin A \cap B$

i.e. $x \notin A$ and $x \notin B$

i.e. $x \in A'$ or $x \in B'$

i.e. $x \in A' \cup B'$

$\therefore x \in (A \cap B)' \Rightarrow x \in A' \cup B'$

$\therefore (A \cap B)' \subseteq A' \cup B'$ -------- (a)

Scanned by TapScanner

Let $x \in A' \cup B'$

i.e. $x \in A'$ or $x \in B'$

i.e. $x \notin A$ and $x \notin B$

i.e. $x \notin A \cap B$

i.e. $x \in (A \cap B)'$

$\therefore \quad x \in A' \cup B' \Rightarrow x \in (A \cap B)'$

$\therefore \quad A' \cup B' \subseteq (A \cap B)' \qquad$ ------ (b)

From (a) and (b), we get $(A \cap B)' = A' \cup B'$

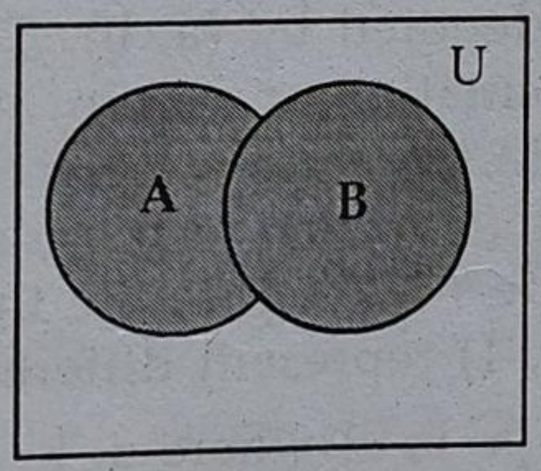
2.8 Venn Diagrams :

The operations on sets can be illustrated with the help of Venn Diagrams. A Venn Diagram is a pictorial representation of sets and is a useful device which expresses the relationship between sets. The universal set U is represented by the set of points inside and on the boundary of a single closed curve, which is usually a rectangle. The subsets of universal set U are usually represented by circles.

We can produce a geometrical interpretation involving sets and operations on sets using Venn diagrams. However, we should note that, Venn diagrams simply illustrate the possibility, but do not prove the truth of a relationship between sets.

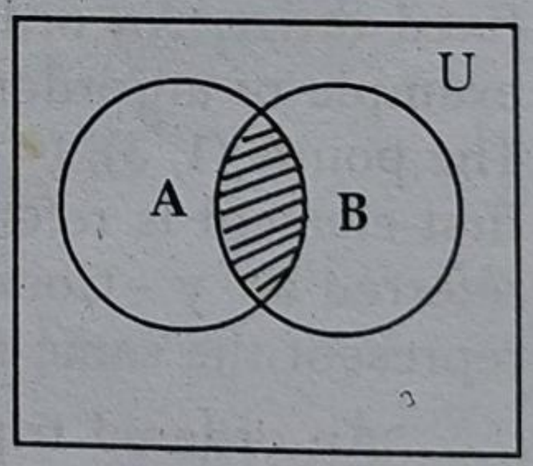
Some examples of Venn diagrams are given below.

(i)



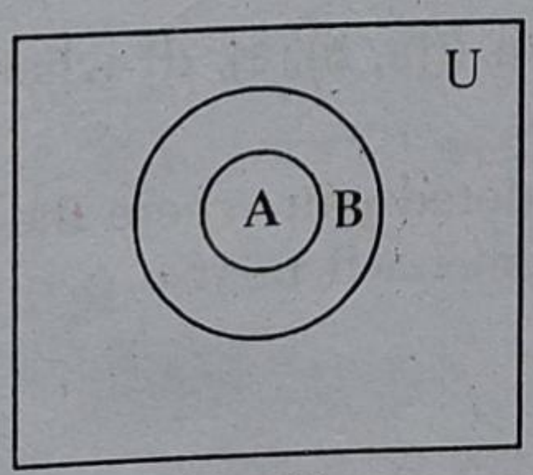
$A \cup B$

(ii)



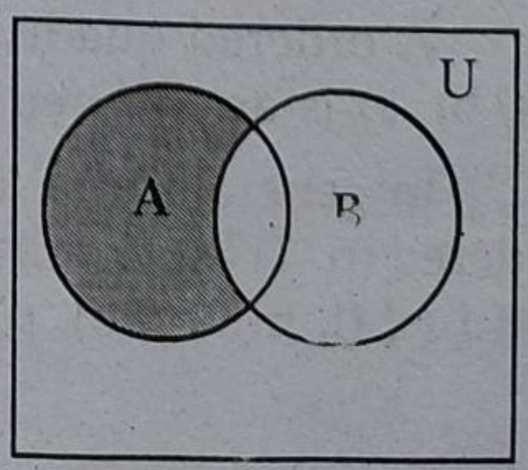
$A \cap B$

(iii)



$A \subset B$

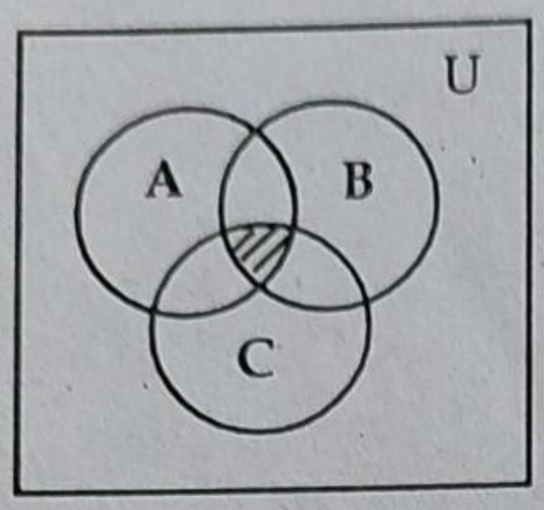
(iv)



A - B

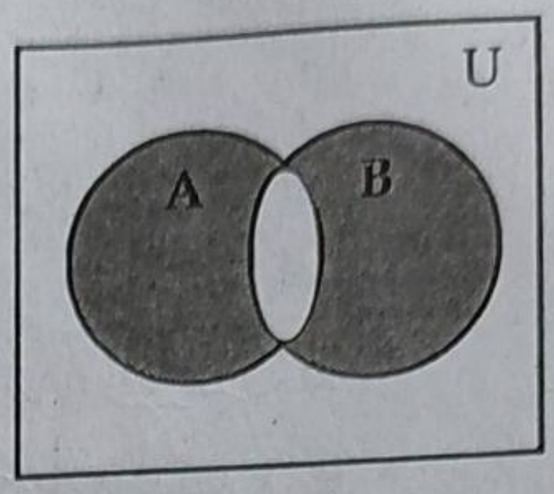
Scanned by TapScanner

(v)



$A \cap (B \cap C) = (A \cap B) \cap C$

(vi)



$A \Delta B = (A - B) \cup (B - A)$

Fig.

2.9 Cardinality :

The number of elements in a finite set A is called the cardinality of A. It is denoted by n(A) or $|A|$.

For example, let $A = \{a, b, c, d, e, x, y, z\}$

Then $n(A) = 8$

Note :

(i) $n(A \cup B) = n(A) + n(B)$, if $A \cap B = \phi$

(ii) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$, if $A \cap B \neq \phi$

(iii) $n(A - B) = n(A) - (A \cap B)$

(iv) $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

2.10 Ordered Pair :

An ordered pair consists of two objects (need not be distinct) in a given fixed order. An ordered pair is usually denoted by (a, b). The most common example of an ordered pair is the representation of a point in two dimension. The points (1, 3), (-1, 4), (2, 0), (3, 4) and (3,1)represent ordered pairs where the first element is referred as x - Coordinate (abscissa) and the second element is referred as y - Coordinate (ordinate). Here we should note that {1, 3} and {3,1} represent the same sets, but (1, 3) and (3, 1) represent different ordered pairs.

An ordered triple is represented by (a, b, c) where a, b, c denote the first, second, third elements respectively of the triple. The ordered triple (a, b, c) can also be written in terms of ordered pair as ((a, b), c).

A ordered quadruple is an ordered pair (((a, b), c), d) where the first element ((a, b), c) is an ordered triple.

In general an ordered n-tuple is an ordered pair where the first element is an ordered (n -1) tuple. For example, a set of 6 elements {l, m, n, o, p, q} can be represented as (((((l, m), n), o), p), q).

Scanned by TapScanner

Cartesian Product :

Let A and B be any two non-empty sets. Then the set of all ordered pairs (a, b) such that a is an element of A and b is an element of B is called the cartesian product of A and B and is denoted by $A \times B$.

i.e. $A \times B = \{ (a, b) \mid a \in A, b \in B \}$

For example, let $A = \{1, 2, 3\}$, $B = \{a, b, c\}$

Then $A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c), (3, a), (3, b), (3, c) \}$

2.11 Properties of Cartesian Products and Sets :

(1) $(A \cap B) \times (C \cap D) = (A \times C) \cap (B \times D)$

(2) $(A - B) \times C = (A \times C) - (B \times C)$

(3) $A \times (B \cup C) = (A \times B) \cup (A \times C)$

(4) $A \times (B \cap C) = (A \times B) \cap (A \times C)$

(5) $(A \cup B) \times C = (A \times C) \cup (B \times C)$

(6) $A - (B \cup C) = (A - B) \cap (A - C)$

(7) $A - (B \cap C) = (A - B) \cup (A - C)$

(8) $(A \cap B) \cup (A \cap B') = A$

(9) $A \cap (B - C) = (A \cap B) - (A \cap C)$

(10) $(A \cap B) - C = (A - C) \cap (B - C)$

Proof : **(1) To prove :** $(A \cap B) \times (C \cap D) = (A \times C) \cap (B \times D)$

We prove that LHS $\subseteq$ RHS and RHS $\subseteq$ LHS

Let $(x, y) \in (A \cap B) \times (C \cap D)$

i.e. $x \in A \cap B$, $y \in C \cap D$

i.e. $x \in A$ and $x \in B$, $y \in C$ and $y \in D$

i.e. $x \in A$, $y \in C$ and $x \in B$, $y \in D$

i.e. $(x, y) \in (A \times C)$ and $(x, y) \in (B \times D)$

i.e. $(x, y) \in (A \times C) \times (B \times D)$

$\therefore$ LHS $\subseteq$ RHS ------- (a)

Let $(x, y) \in (A \times C) \cap (B \times D)$

i.e. $(x, y) \in A \times C$ and $(x, y) \in B \times D$

i.e. $x \in A$, $y \in C$ and $x \in B$, $y \in D$

i.e. $x \in A$ and $x \in B$, $y \in C$ and $y \in D$

Scanned by TapScanner

i.e. $x \in (A \cap B),\ y \in (C \cap D)$

i.e. $(x, y) \in (A \cap B) \times (C \cap D)$

$\therefore$ $(x, y) \in (A \times C) \cap (B \times D) \Rightarrow (x, y) \in (A \cap B) \times (C \cap D)$

$\therefore$ RHS $\subseteq$ LHS -------- (b)

From (a) and (b), we get $(A \cap B) \times (C \cap D) = (A \times C)\ (B \times D)$

(2) To prove : $(A - B) \times C = (A \times C) - (B \times C)$

We prove that LHS $\subseteq$ RHS and RHS $\subseteq$ LHS

Let $(x, y) \in (A - B) \times C$

i.e. $x \in A - B, y \in C$

i.e. $x \in A$ and $x \notin B, y \in C$

i.e. $x \in A, y \in C$ and $x \notin B, y \in C$

i.e. $(x, y) \in (A \times C)$ and $(x, y) \notin (B \times C)$

i.e. $(x, y) \in (A \times C) - (B \times C)$

$\therefore$ $(x, y) \in (A - B) \times C \Rightarrow (x, y) \in (A \times C) - (B \times C)$

$\therefore$ LHS $\subseteq$ LHS -------- (a)

Let $(x, y) \in (A \times C) - (B \times C)$

i.e. $(x, y) \in A \times C$ and $(x, y) \notin B \times C$

i.e. $x \in A, y \in C$ and $x \notin B, y \notin C$

But $y \in C$

i.e. $x \in A,\ y \in C$ and $x \notin B$

i.e. $x \in A$ and $x \notin B, y \in C$

i.e. $x \in A - B, y \in C$

i.e. $(x, y) \in (A - B) \times C$

$\therefore$ RHS $\subseteq$ LHS -------- (b)

From (a) and (b), we get $(A - B) \times C = (A \times C) - (B \times C)$

Scanned by TapScanner